PLACETRACK: CAMPUS PLACEMENT SYSTEM

Project Completion Report, Executive Summary, Abstract & Architecture

STATUS: 100% COMPLETE

Target Domain:	Platform Architecture:	Target & Tested Device:	Code Quality Audit:
Higher Education / Recruitment	**Flutter (Dart 3.x) Cross-Platform**	**Vivo I2221 (Android 16 / API 36)**	**0 Errors / 0 Warnings (Passed)**

1. PROJECT ABSTRACT

Campus placement operations in higher education institutions typically struggle with fragmented communication, manual candidate eligibility verification, delayed stage feedback, and opaque recruitment metrics. PlaceTrack is an enterprise-grade campus placement and career management system built with Flutter and Dart. The platform creates a unified digital bridge connecting students, training coordinators, and Training & Placement Officers (TPOs). Core capabilities include automated eligibility filtering (CGPA, active backlogs, department matching), multi-stage drive tracking with exact round timestamps, real-time dynamic placement analytics, and an interactive celebration physics engine. Tested and deployed directly on physical Android 16 hardware, PlaceTrack accelerates recruitment turnaround, eliminates manual spreadsheet tracking, and delivers complete institutional transparency.

2. EXECUTIVE SUMMARY & ACHIEVED OBJECTIVES

- **Automated Eligibility Engine:** Instant verification of CGPA and backlog thresholds prevents unauthorized registrations.
- **End-to-End Recruitment Pipeline:** Orchestrates applicant enrollment, multi-round technical assessments, and final offer letters.
- **Dynamic Real-Time Analytics:** Live computation of placement ratios, average/highest CTC, and department conversion funnels.
- **Interactive Morale Boosting:** Foreground rocket and particle burst simulation celebrating student offer letter unlocks.

3. ARCHITECTURE & SYSTEM WORKFLOW

[STUDENT PORTAL]
1. View Eligible Drives
2. One-Click Enrollment
3. Resume PDF Upload
4. Mock Tests & Preparation
5. Offer Letter & Badge

SYNC

[CENTRAL DATA SERVICE]
- PlacementDataService Singleton
- ChangeNotifier Reactive State
- Real-Time Metrics Engine
- Exact Timestamps Audit Log
- Offline-Ready Architecture

SYNC

[TPO / ADMIN PORTAL]
1. Create & Schedule Drives
2. Configure Multi-Rounds
3. Round-by-Round Evaluation
4. Confirm Placed Candidates
5. Executive Analytics Funnel

4. TOOLS & TECHNOLOGIES USED

Category	Tool / Library	Implementation & Architectural Purpose
Framework & Core	Flutter SDK (3.x) & Dart 3.13+	Single codebase rendering high-performance native UI on Android, Windows, and Web.
UI & Theming	Material 3 & PlaceTrackTheme	Unified design tokens supporting dynamic, high-contrast Light and Dark mode styling.
State Management	ChangeNotifier Singleton Service	Centralized reactive state synchronizing real-time analytics, drives, and student records.
Graphics Engine	CustomPainter & Physics Engine	Simulates realistic rocket trails, multi-color explosive stars, and radial shockwaves.
Document Engine	pdf (3.13.1) & printing (5.15.1)	Generates authentic corporate offer letters and comprehensive PDF completion reports.
Build & Deployment	Gradle 9.3.1 & Android SDK API 36	Compiles native ARM64 binaries; direct streamed installation via ADB to hardware devices.

5. WORK COMPLETION AUDIT & MODULE BREAKDOWN

Module Name	Implemented Capabilities	Status
Student Portal	Live drive cards, enrollment, resume manager, round status tracking, offer downloads.	**100% Complete**
TPO Drive Manager	Drive builder, date scheduler, multi-rounds creator, CGPA filters, enrollment lists.	**100% Complete**
Multi-Round Evaluation	Student qualification/rejection dialogs, batch actions, exact timestamp logging.	**100% Complete**
Real Dynamic Analytics	Replaced all demo values with computed CTC, department statistics, and hiring funnel.	**100% Complete**
Explosive Celebration	Foreground fireworks particle simulation with Placed On timestamp badge.	**100% Complete**
Aptitude & DSA Quizzes	Timed mock tests with category selection, immediate score evaluation & explanations.	**100% Complete**
High-Contrast Dark Mode	Theme-aware semantic colors guaranteeing zero contrast loss on all OLED/LCD screens.	**100% Complete**

6. PROJECT STATUS & PHYSICAL DEVICE DEPLOYMENT

☒ **Static Code Quality Audit: PASSED**
Validated with `dart analyze lib` yielding 0 issues across all 25+ source files.

☒ **Installation Status: SUCCESSFUL**
Streamed install verified via ADB (`adb install -r build/.../app-debug.apk`).

☒ **Android Target Device: Vivo I2221 (ARM64)**
Running latest Android 16 (API Level 36) with full hardware acceleration enabled.

☒ **Production Readiness: 100% READY**
All modules tested end-to-end; documentation artifacts compiled in /docs directory.

7. CONCLUSION & VERIFICATION SIGN-OFF

PlaceTrack has met all designated system specifications, functional criteria, and quality standards. The system is certified complete, fully functional, and ready for institutional deployment.

COMPLETION STATUS
APPROVED / VERIFIED